blue color; and when the temperature is about right, the LED will change to green, with all of the colors possible between the two extremes. To make this work, we are going to need more complex color-mixing than we have done previously with seven colors. Rather than make our lives more difficult than they need to be, we will replace our RGB LED with the BlinkM MinM smart LED for this project. The BlinkM is an RGB LED, like we have previously used, that has been coupled with a small Arduino-like microcontroller with some secret software, or firmware, that has been preloaded on the device. This means that the BlinkM is capable of handling all of the color mixing for us using, at least in our example, the Hue Saturation Brightness or HSB color model shown in Figure 7-1.

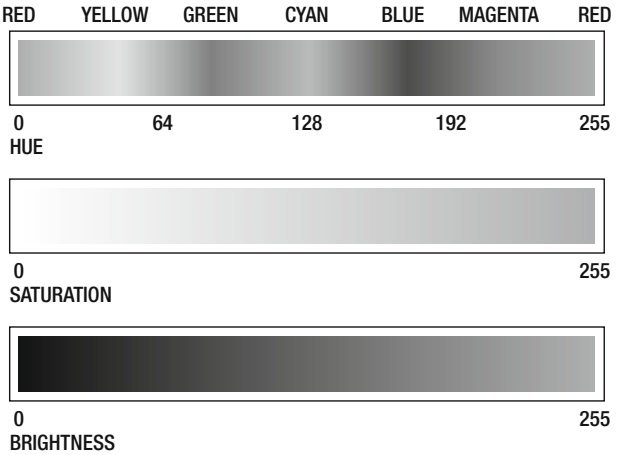

Figure 7-1. *HSB color model*

Again, we have black-and-white images of color models, but if you remember back to Chapter 2 where we discussed the RGB color wheel, HSB handles color values by specifying the colors hue with a value from 0 to 255, beginning with red and rotating through all the colors in the rainbow until we get back to red on the other end. We can also specify a color's saturation (the vividness of the color) and the color's brightness (the lightness or darkness of the color). This will work nicely in our project code because we can keep the saturation and brightness at the same level while fading through multiple colors with red at 0 on one end and blue at 170 on the other.

Hooking It Up

This basic circuit combines the BlinkM MinM smart LED with the TMP36 temperature sensor that comes in a package similar to our transistor from the last project, but with a different pin connection. It's still super-easy to hook up and, once connected to +5 volts and ground, the output pin will provide a linear reading that corresponds to the temperature measured in Celsius, as shown in Figure 7-2.